********** QUESTIONS AND SOLUTIONS **********

Information for Candidates

Numerical answers

- Numerical answers should be given as fully evaluated decimal values and not as unevaluated arithmetic expressions.

Notation

The following notation is used in this paper:

- The voltage waveform at node X in an AC circuit is denoted by $x(t)$ while the phasor voltage is denoted by X.

- Component and source values are generally given in Ohms, Farads, Henrys, Volts or Amps with the unit symbol omitted. Where an imaginary number is specified, it represents the complex impedance or phasor value.

- Times are in seconds unless otherwise stated.

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The Questions

1. a) For the circuit shown in Figure 1.a, calculate the voltage at node *X*.

[6]

KCL at X gives : $\frac{X-10}{10} - 1 + \frac{X}{15} + \frac{X+3}{9} \Rightarrow X = 6$

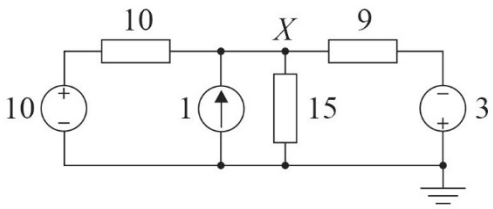


Figure 1.a

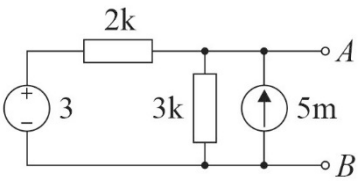
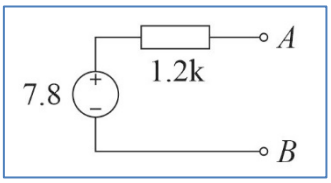


Figure 1.b

b) Draw the Thévenin equivalent of the network shown in Figure 1.b, assigning values to the elements.

[5]

Using superposition for the open-circuit voltage: $V_{AB} = \left(\frac{3}{5}\right) \times 3 + \left(\frac{6}{5}\right) \times 5 = 7.8\ \text{V} = V_{Th}$
And, by inspection, $R_{Th} = 2k \parallel 3k = 1.2k$



c) For the circuit shown in Figure 1.c, give an expression for the voltage *Z* in terms of the voltages U_1 and U_2.

[6]

The voltage at the + input of the opamp is (from potential divider rule): $V+= \left(\frac{10}{15}\right) \times U_1 + \left(\frac{5}{15}\right) \times U_2$, and the gain from the + input to the output is 3. Combining these we get:

$$Z = 2U_1 + U_2$$

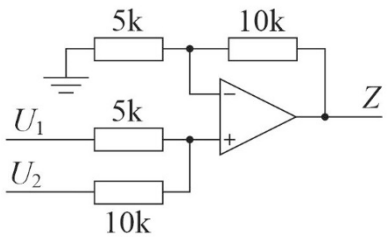


Figure 1.c

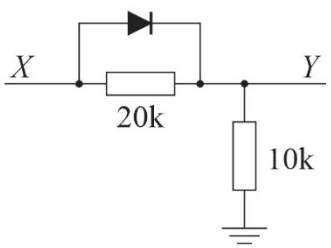


Figure 1.d

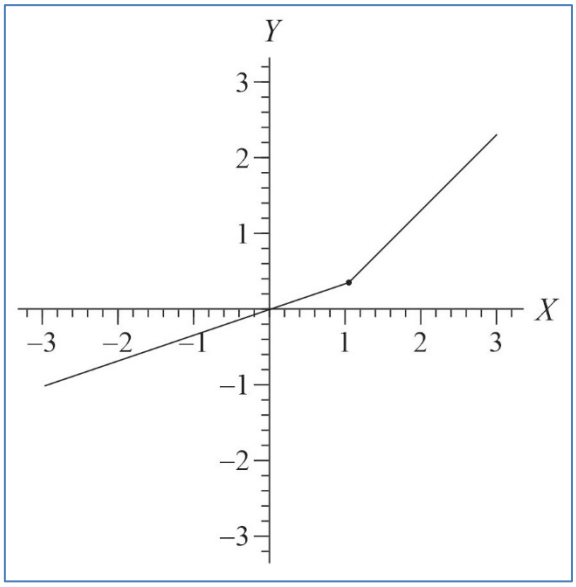
d) For the circuit shown in Figure 1.d, draw a dimensioned sketch showing the variation of voltage Y with voltage X, covering the range $-3 \leq X \leq 3$. You should assume that the diode has a forward voltage of 0.7 V but is otherwise ideal.

[6]

The diode will be in conduction whenever the voltage that would be dropped across the 20k resistor in the absence of the diode would be >0.7 V. So, the threshold for conduction in the diode is $X = \left(\frac{3}{2}\right) \times 0.7 = 1.05$ V.

For $X \leq 1.05$, the circuit is just a potential divider with $Y = X/3$.

For $X > 1.05$, the diode will clamp the voltage between X and Y at 0.7 V.



e) The circuit in Figure 1.e is biased such that 1 mA of collector current is flowing. Calculate the small signal input resistance, output resistance and gain of the circuit.

[5]

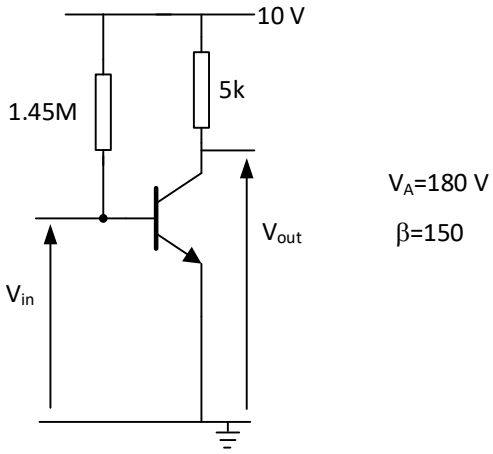


Figure 1.e

If 1mA is flowing, the small signal parameters of interest are:

$g_m=I_C/V_T=1m/25m=40$ mS
$r_{be}=\beta/g_m=150/0.04=3.75k\Omega$
$r_o=E_A/I_C=180/1m=180k\Omega$

The input resistance is therefore 1.45M//3.75k=3.74kΩ
The output resistance is r_o//5k=4.87 kΩ.
*The gain is therefore $-g_m$*4.87k=-194.8*

f) Calculate the drain current in the MOSFET of Figure 1.f.

[6]

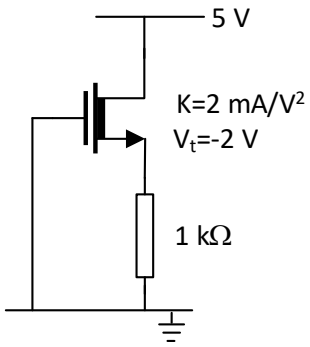


Figure 1.f

Assuming the MOSFET is saturated, the current in the MOSFET is:

$I_D=k(V_{GS}-V_t)^2=-V_{GS}/1k$
Therefore: $2(V_{GS}+2)^2=-2$
So $(V_{GS}+1)(V_{GS}+3)=0$
If V_{GS}=-3V, the MOSFET is off, but this would require VS=3V and hence requires current in the resistor. Therefore, V_{GS}=-1 V, and so I_D=1mA.

g) The circuits in Figure 1.g are unity voltage gain amplifiers. By sketching waveforms, compare and contrast their performance. What function do resistors R3 and R4 perform?

[6]

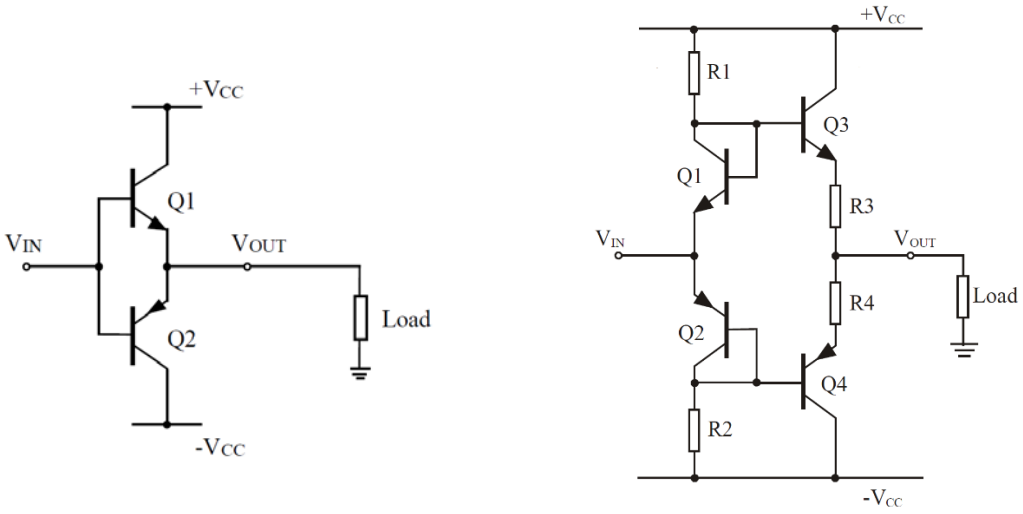
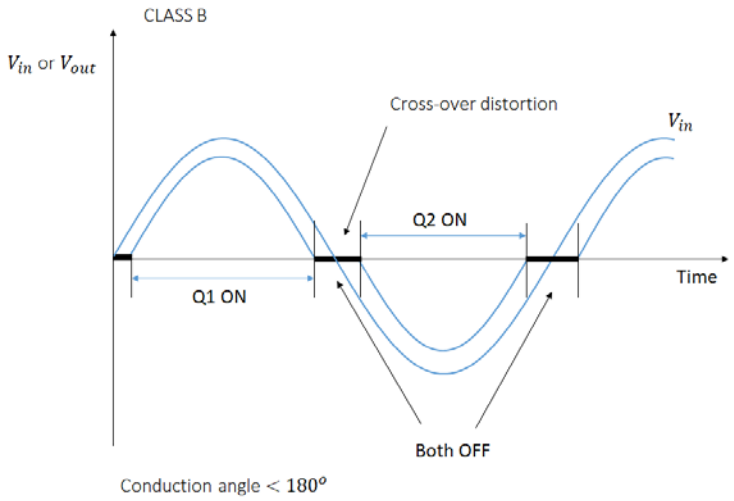


Figure 1.g

The left diagram is known as a class B amplifier and the right is a class AB amplifier. These circuits provide current gain and are thus often used as output stages. The class B amplifier suffers from crossover distortion (when V_{IN} has a magnitude less than 0.7V, neither transistor is on and the output remains at 0 V.) This is shown below:



The class AB circuit removes the cross over distortion by biasing the output transistors, by adding or subtracting 0.7 V from the input voltage. The resistors R3 and R4 limit the quiescent current in output transistors Q3 and Q4 (i.e. limiting shoot through when V_{IN}=0V) and prevent cross conduction when V_{IN}!=0..

2. Figure 2.1 shows an active band-pass filter comprising an opamp in inverting amplifier configuration with an LCR network in the feedback path.

a) Explain briefly, with reference to the individual circuit elements, why the circuit acts as a band-pass filter.

[4]

The general expression for the gain of a non-inverting amplified with input and feedback impedances Z_i and Z_f is $-Z_f/Z_i$. In this case Z_i is fixed, and the frequency dependence in the gain comes purely from Z_f. This impedance becomes small at low frequencies because of the inductor and at high frequencies because of the capacitor. $|Z_f|$ is maximum at the resonance frequency where the effects of the L and the C cancel and $Z_f = R$.

b) Obtain an expression for the transfer function $H(j\omega) = \frac{Y}{X}$ of the filter, and hence show that the centre frequency and quality factor are given by:

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad ; \quad Q = \frac{R_2}{\omega_0 L}$$

[8]

The feedback impedance $Z_f = \left(\frac{1}{R_2} + j\omega C + \frac{1}{j\omega L}\right)^{-1}$, so the transfer function is:

$$H = -\frac{1}{\left(\frac{1}{R_2} + j\omega C + \frac{1}{j\omega L}\right)}\frac{1}{R_1} = -\frac{1}{R_1}\frac{j\omega L}{LC(j\omega)^2 + \frac{j\omega L}{R_2} + 1}$$

Comparing with the standard form, we see that $LC = 1/\omega_0^2$ and $\frac{2\zeta}{\omega_0} = L/R_2$, from which the required results follow.

c) If the filter is required to have a centre frequency of 10 kHz, a 3 dB bandwidth of 500 Hz a peak gain of 0 dB, and an input impedance of 1 kΩ, what values should be used for the four passive components?

[8]

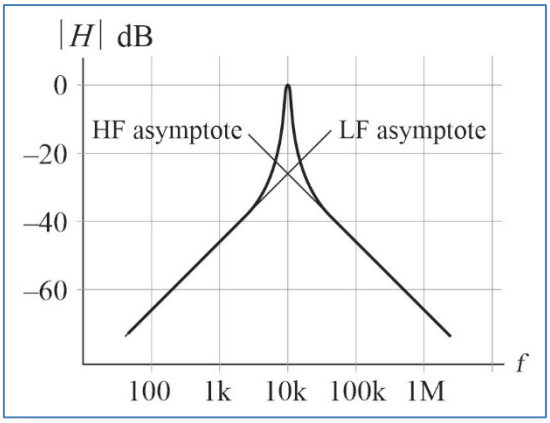
The input impedance of the filter is R_1, so we know $R_1 = 1$ kΩ.

The peak magnitude of the gain is R_2/R_1 and we need this to be unity, so $R_2 = 1$ kΩ.

The Q needs to be 10000/500 = 20*, so the inductor value is $L = \frac{R_2}{\omega_0 Q} = 796$ µH.*

Finally, we can calculate the capacitor value using $C = \frac{1}{L\omega_0^2} = 318$ nF.

d) Assuming the component values you calculated in part c), draw a dimensioned sketch showing the variation with frequency of $|H|$ together with the low- and high-frequency asymptotes.



[6]

e) It is suggested that the same frequency response can be achieved with the filter shown in Figure 2.2. Is this true and, if so, what are the potential advantages of the active circuit over the passive one?

[4]

The circuit in Figure 2.2 is also a 2nd order band-pass filter with unity gain at the centre frequency, so yes, it is true. The main advantage of the active circuit are that the input impedance is frequency independent; it is also higher in the pass-band than that of the passive circuit (assuming same specs and same L & C values). Low output impedance could also be mentioned, although the passive circuit will also have this (again assuming same specs and same L & C values).

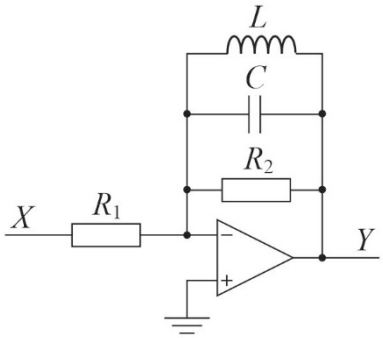


Figure 2.1

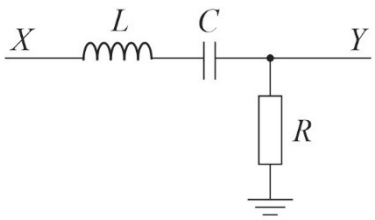


Figure 2.2

3. Figure 3.1 shows a differential amplifier. All transistors are identical.

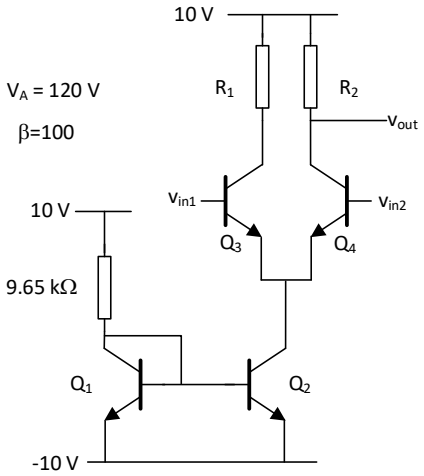


Figure 3.1

a) Calculate values of R_1 and R_2 to bias V_{out} at 0 V with zero differential input.

[2]

The current mirror that provides the tail current has a reference current of: (20-0.7)/9.65k=2 mA.

This means that the emitter currents in Q_3 and Q_4 is 1 mA each. To drop 10 V across R_1 and R_2 with 1 mA flowing requires R_1=R_2=10 kΩ..

b) Calculate the quiescent power consumption of the circuit

[3]

*The quiescent power can be calculated by looking at the current drawn from the power supply, as being 2 mA (for the collector current of Q_1) plus 1 mA for each collector of Q_3 and Q_4, giving a total quiescent power of 4 mA*20 V=80 mW.*

c) Draw a small signal model of the circuit in differential mode and calculate the single-ended differential gain.

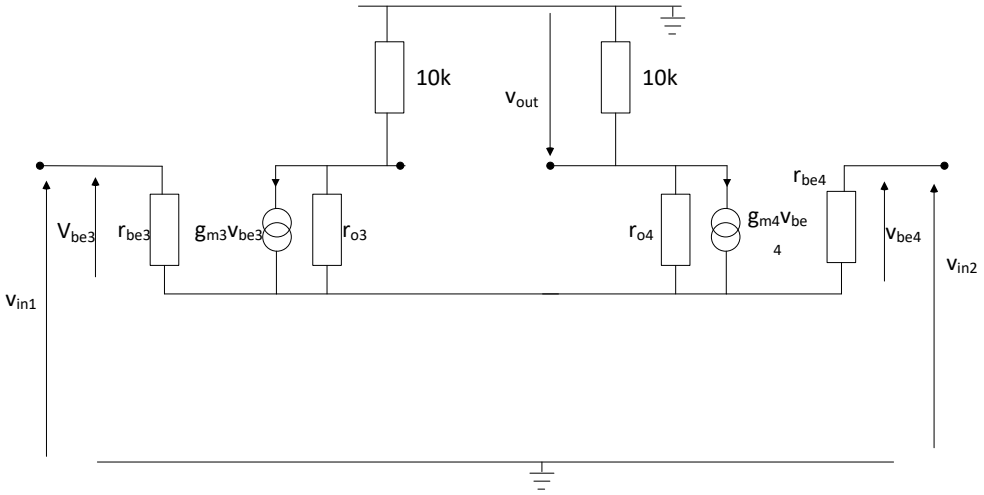
[5]

The small signal model is shown overleaf.

The output resistance of transistors Q_3 and Q_4 is V_A/I_c=120/1mA=120kΩ.
g_m for Q_3 and Q_4 is 1mA/25mV=0.04 S

The single ended differential mode gain is therefore:
$A_d=g_m(r_o//10k\Omega)/2$=40mS(120k//10k)/2=40m*(9.23k)/2=184.6*

(Note that the valu is positive, assuming $V_{in}=V_{in1}-V_{in2}$)



d) State and explain the common mode voltage range

[4]

The CMVR is the range for over which common mode inputs can be applied whilst keeping the common mode of the amplifier low. The CMVR is for input voltages (in common mode) between -8.6 V (to prevent the output transistor of the current mirror from entering the saturation mode. The maximum value of the input voltages in common mode is 0 V (to prevent transistors Q_3 and Q_4 saturating.

It is now proposed to replace R_1 and R_2 with current mirror outputs, as shown in Figure 3.2.

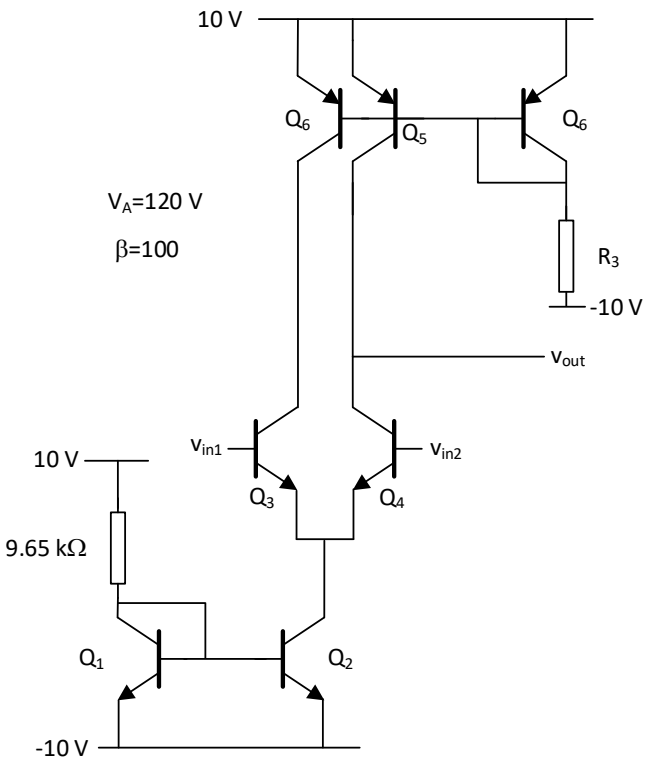


Figure 3.2

e) Calculate the value of R_3 to make the collector current of Q_3 and Q_4 the same as it was in Figure 3.1 and calculate the quiescent power consumption of the circuit of Figure 3.2.

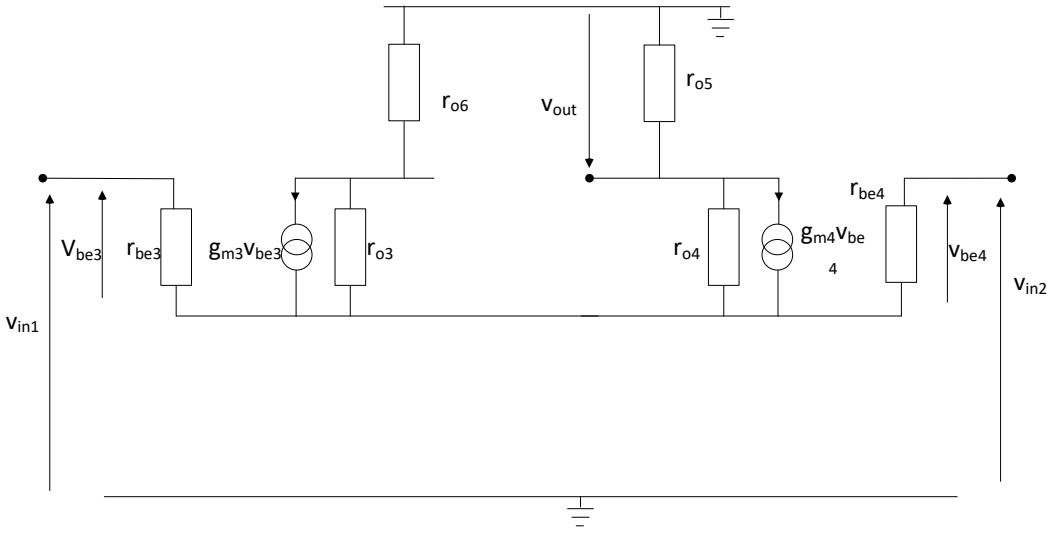
[4]

We need to set the output of the new current mirror to 1 mA, hence R2 should be set to 19.3 kΩ. The power consumption of the new circuit is now:
*5 mA*20 V=100 mW.*

f) Calculate the common-mode voltage range and then, by drawing a new small signal model, calculate the differential mode gain.

[8]

The small signal model is shown below:



The differential mode gain is now $g_m*(r_{o4}//r_{o5})/2=40m*60k/2=1200$.

The lower limit for the CMVR is the same as the previous circuit, i.e. -8.6 V. The upper limit is now 9.3 V (to prevent Q_5 and Q_4 from saturating.

g) Explain what happens to the output voltage if, due to component tolerances, the output of the current mirror that has replaced R_1 and R_2 is higher or lower than half of the tail current. Assume $v_{in1}=v_{in2}=0$ V.

[4]

If the currents between the two mirrors are mismatched, then the output voltage will quickly saturate to a high or a low value, depending on which way the currents are mismatched. In the case that the lower current mirror has a greater output current than the upper current mirror, for a common-mode input, the output voltage will saturate towards around 0.2 V (as Q_3 and Q_4 will go into saturation, so that the voltage drop across the collector-emitter of Q_5 and Q_6 would be as large as possible. If the upper current mirror has a greater output current than the lower one, the output voltage will move to around 9.8 V as Q_5 saturates.